Ricardo **Vilela de Godoy**

PhD Mechatronics Engineer · Postdoc researcher

São Carlos, Brazil

(+55) 17991523513 | ricardo.godoy@usp.br | www.ricardovgodoy.com |

Ricardo Vilela de Godoy | Ricardo Vilela de Godoy | 0000-0002-5323-9299



Summary

I am a robotics researcher focused on **Physical AI**, integrating learning-enabled robot capabilities into real-world **inspection**, **automation**, and **assistive** frameworks. I am currently a Postdoctoral Researcher at the University of São Paulo (USP), working on the development of robotic frameworks for inspection and automation, with an emphasis on **manipulation and locomanipulation**. I received my PhD at the University of Auckland, where I developed **human–robot interfaces and shared-control** methods to combine **biosignals** (e.g., EMG) and **wearable devices** with external sensing (e.g., **motion capture and vision**) to decode discrete and continuous human motion for intuitive **control of robotic and prosthetic** devices. During my MSc at USP, I worked on **EEG-based epileptic seizure prediction**. My primary interests are in robotics, sensing, and learning, with the goal of developing systems that are **robust for real-time control in real-world settings** across **industrial, medical, and daily-life applications**.

Professional Experience

University of São Paulo
São Carlos, Brazil

Postdoctoral Research Fellow — *Sep. 2024 - Present*

- Mechatronics engineer working on the development of robotic frameworks for inspection and maintenance in oil platforms
- Technical leader of a team of 30-50 members and experienced in arranging international collaborations between universities
- Research in robotics, locomanipulation frameworks, and machine learning applications

Faculty of Health Sciences, Albert Einstein Hospital
São Paulo, Brazil

Assistant Professor — *Jan. 2025 - Jul. 2025*

- Assistant professor in the Biomedical Engineering Bachelor's program and in the Postgraduate Degree in Bioengineering Applied to Health.

New Dexterity Research Group, The University of Auckland
Auckland, New Zealand

Graduate Research Associate — *Dec. 2021 - Aug. 2024*

- Mechatronics engineer working on the development of novel human-machine interface solutions
- Development of novel bionic devices and deep learning algorithms

Research associate - Collaboration with Acumino (USA), part-time — *Dec. 2021 - Aug. 2024*

- Data collection and analysis of grasping and manipulation strategies using wearable human machine interfaces
- Development of machine learning-based algorithms for automated annotation of videos

Research associate - Collaboration with Prowood Limited (NZ), part-time — *Sep. 2022 - Mar. 2023*

- Funded by the 2022/23 R&D Experience Grants from Callaghan Innovation New Zealand's Innovation Agency
- Development of an automated framework for assembling beehive frames

University of São Paulo
São Carlos, Brazil

Graduate Research Associate — *Jul. 2019 - Jul. 2021*

- Mechatronics engineer working on the development of novel deep learning and deep reinforcement learning techniques
- Member of the robotic surgery group
- Research in neurology, epilepsy, machine learning, and brain-machine interface

University of São Paulo
São Carlos, Brazil

Undergraduate researcher - Fundação de Apoio à Física e à Química (FAFQ) — *Jul. 2016 - Dec. 2019*

- Engineering undergraduate researcher responsible for the development and implementation of an algorithm based on Dynamic Movement Primitives in a robotic arm. I also worked on the simulation of the autonomous vehicle using V-REP and ROS

MULTITTECH Engineering
São Carlos, Brazil

Engineering Intern — *Jan. 2019 - Jul. 2019*

- Intern in the modelling and simulation of dynamic systems group

Education

The University of Auckland — *Auckland, New Zealand*

PhD in Mechanical and Mechatronics Engineering — *Dec. 2021 - Aug. 2024*

- Thesis on analysis and development of novel human-machine interfaces for the control of bionic devices
- Research in robotics, human-machine interfaces, and machine learning techniques

University of São Paulo — *São Carlos, Brazil*

Master in Mechanical Engineering — *Jan. 2020 - Jul. 2021*

- Thesis Title: Epileptic Seizure Prediction using Deep Learning Techniques
- Research in machine learning techniques, brain-computer interfaces, neuroscience, neuroimaging, and robotics

University of São Paulo — *São Carlos, Brazil*

Bachelor in Mechatronics Engineering — *Feb. 2015 - Dec. 2019*

- Senior Thesis Title: Comparison of Deep Reinforcement Learning Control Methods of Autonomous Robot in a Competition Task

Competitive fellowships & funded research positions

- **FAPESP Postdoctoral Fellowship**, São Paulo Research Foundation (FAPESP), Postdoctoral Fellow, 2026–present.
- **Postdoctoral fellowship (industry-funded)**, Petrobras project, Postdoctoral Fellow, 2024–present.
- **PhD scholarship (industry-funded)**, Acumino, Graduate Research Associate, 2021-2024.
- **R&D Experience Grant**, Callaghan Innovation New Zealand's Innovation Agency, 2022-2023.
- **MSc and BSc scholarships**, Fundação de Apoio à Física e à Química, 2019-2021.

Research supervision and leadership

- Co-supervision: 3 PhD, 1 MSc, 6 BSc. Full list, alongside research topics and thesis titles, available at `https://www.ricardovgodoy.com/supervision/`.
- Leadership: Petrobras Project. Technical leader of a team of 30-50 members coordinating project deliverables and scientific outputs across different areas, including manipulation, locomotion, navigation, and perception.

Honors and Awards

2026	**Best Poster Presentation Award**, The 1st International Workshop on Industrial Applications of Robot Learning at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	*Pittsburgh, US*
2025	**Best Paper Award**, IEEE Latin American Robotics Symposium (LARS 2025)	*Monterrey, Mexico*
2009-2014	**1 gold medal, 2 bronze medals, and 1 honorable mention**, Paulista Physics Olympiad (OPF)	*Brazil*
2007-2014	**1 gold medal and 6 silver medals**, Brazilian Astronomy Olympiad (OBA)	*Brazil*
2011	**Honorable mention**, Brazilian Physics Olympiad (OBF)	*Brazil*
2010	**Gold medal**, Brazilian Robotics Olympiad (OBR)	*Brazil*

Teaching Experience

Invited Lecturer - University of São Paulo (USP) — *São Carlos, Brazil*

Neural Networks Course: Postgraduate Degree in Mechanical Engineering — *2026*

- I was invited to deliver guest lectures on Transformer architectures
- Provided graduate students with both theoretical grounding and concrete examples from a state-of-the-art research application

Assistant Professor - Faculty of Health Sciences, Albert Einstein Hospital — *São Paulo, Brazil*

Specialization: Postgraduate Degree in Bioengineering Applied to Health — *2025*

- Emphasized a multidisciplinary, technology-driven approach to tackling complex healthcare challenges
- Focus on development of machine learning-based projects, applied electrical circuits and electronics, and signal processing

Course: Processing of Medical Signals and Images — *2025*

- Course taught for the Biomedical Engineering Bachelor's program
- Biosignal physiology and signal/image processing for medicine and robotics applications
- Reinforced concepts with hands-on Python labs using core scientific libraries

Graduate Teaching Assistant - The University of Auckland — *Auckland, New Zealand*

MECHENG 730/736: Biomechatronics Systems — *2022-2024*

- Explore mechatronic principles and techniques for measuring and manipulating biological systems
- Design of biomechatronics systems, human biomechanics, and motion control

MECHENG 700: Part IV (Final Year) Research Project *2022-2024*

- Development of research projects on a topic of interest for Mechanical and Mechatronics Engineers

Workshops, Tutorials, and Talks

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) — *Pittsburgh, USA*
Two poster presentations: The 1st International Workshop on Industrial Applications of Robot Learning (**Best Poster Presentation Award**); and Reproducible Benchmarking of Robotic Grasping and Manipulation: From Advanced AI to Generalized Humanoid Intelligence — *Sep. 2026*
Poster: HiveBoard: An Open, Modular, 3D-Printed Benchmark of Industrial Mechanisms for Robotic and Prosthetic Manipulation

IEEE International Conference on Robotics and Automation (ICRA) — *Vienna, Austria*
Poster presentation in the Workshop on Manipulation Robustness: Towards Human-Level Robustness under Real-World Challenges — *Jun. 2026*
Poster: Viewpoint-Agnostic Grasp Pipeline using VLM and Partial Observations

XXII Sematron - University of São Paulo — *São Carlos, Brazil*
Talk: Toward Physical AI - Integrating Embodiment, Control, and Learning in Robotic Systems — *May 2026*

IEEE International Conference on Advanced Robotics (ICAR) — *San Juan, Argentina*
Workshop organizer: Robotic Fault-Detection Strategies for Complex Industrial Contexts — *Dec. 2025*

University of São Paulo — *São Carlos, Brazil*
Invited talk: Development of Human-Machine Interfaces for the Teleoperation Control of Robotic Manipulators — *Jun. 2025*

Hospital Albert Einstein — *São Paulo, Brazil*
Invited talk: Human-Machine Interfaces: Applications of Biological Signals in Robotics — *Aug. 2024*

University of Waikato — *Hamilton, New Zealand*
Invited talk: Analysis and Development of Human-Machine Interfaces for the Control of Robotic and Bionic Devices — *Jan. 2024*

IEEE International Conference on Automation Science and Engineering (CASE) — *Auckland, New Zealand*
Tutorial: Biosignal-Based Design Approaches for the Development of Human-Machine Interfaces for Shared Control of Computer Applications and Robotic Devices — *Aug. 2023*

Service

IEEE Robotics and Automation Letters (RA-L) — *IEEE*
Associate Editor — *2026 - Today*
Area: Robot Learning

IEEE International Conference on Robotics & Automation (ICRA 2027) — *IEEE*
Associate Editor — *2026 - 2027*
Area: Human-Centered Robotics and Automation

Peer reviewer — *International*
Journals — *2020 - Today*

- Nature Portfolio: Scientific Reports, npj Robotics
- IEEE Robotics and Automation Letters (RA-L), IEEE Journal of Biomedical and Health Informatics, IEEE Access
- Elsevier: Neural Networks, Computers in Biology and Medicine, Computational and Structural Biotechnology Journal, Journal on Multimodal User Interfaces
- Springer Nature: Archives of Computational Methods in Engineering, Journal of Bionic Engineering, Journal of NeuroEngineering and Rehabilitation, Journal of Medical Systems, Journal of Intelligent & Robotic Systems, Signal Image and Video Processing, Discover Artificial Intelligence, Journal of Machine Learning and Cybernetics, Pattern Analysis and Applications, Engineering Applications of Artificial Intelligence, Journal of Control, Automation and Electrical Systems
- Frontiers in Virtual Reality

Conferences — *2020 - Today*

- Conference on Robot Learning (CoRL), IEEE International Conference on Intelligent Robots and Systems (IROS), IEEE International Conference on Robotics and Automation (ICRA), IEEE Conference on Biomedical Robotics and Biomechatronics (BioRob), IEEE International Conference on Advanced Robotics (ICAR), and member of the Program Committee on the 2026 IEEE Latin American Robotics Symposium (LARS)

IEEE RAS Young Reviewer Program (YRP) — *IEEE*
Senior Reviewer (SR) — *2026*

São Paulo Research Foundation (FAPESP) — *São Paulo, Brazil*
Ad hoc scientific reviewer — *2026 - Today*

The University of Auckland — *Auckland, New Zealand*
Liaison of the Mechatronics Laboratory at the University of Auckland — *2023 - 2024*

IEEE International Conference on Automation Science and Engineering (CASE) — *Auckland, New Zealand*
Volunteer at the 2023 International Conference on Automation Science and Engineering — *Aug. 2023*

Museum of Transport and Technology Auckland (MOTAT) STEM Fair 2023 — *Auckland, New Zealand*
Specialist exhibitor at the 2023 MOTAT STEM Fair — *Apr. 2023*
- Demonstrations and displays of my research to engage the next generation of kids to consider careers in STEM

2022 Conference on Robot Learning — *Auckland, New Zealand*
Audio Visual Equipment Chair at the 2022 Conference on Robot Learning — *Dec. 2022*

2022 World Robot Olympiad — *Auckland, New Zealand*
Judge in the 1st New Zealand Finals of the World Robot Olympiad — *Oct. 2022*

IEEE-RAS Student Chapter — *São Carlos, Brazil*
Co-founder and vice president of the RAS Chapter at the University of São Paulo — *2019 - 2021*

Publications

Journal Publications

B. Guan, Z. Wang, **R. V. Godoy**, M. Owen, and M. Liarokapis, "Combining Augmented Reality with Semi-autonomous, Lightmyography Based Control to Improve Usability of Prostheses", in *IEEE Access*, 2026.

T.H. Segreto, J. Negri, P. H. Polegato, J. M. H. Pinheiro, **R. V. Godoy**, and M. Becker, "A Leaf-Level Dataset for Soybean-Cotton Detection and Segmentation", in *Scientific Data*, 2026.

J. M. H. Pinheiro, G. N. Herrera, A. D. Santos, L. B. R. Fernandes, A. D. Santos, **R. V. Godoy**, E. A. B. Almeida, M. A. C. Vieira, A. M. Penteado-Dias, and M. Becker, "Descriptor: Parasitoid Wasps and Associated Hymenoptera Dataset (DAPWH)", in *IEEE Data Descriptions*, 2026.

M. V. da Silva, **R. V. Godoy**, J. Negri, G. J. G. Lahr, R. Bezerra, and M. Becker, "From Perception to Assistance: Open-Vocabulary Shared Autonomy for Robotic Manipulation", in *IEEE Robotics and Automation Letters (RA-L)* (under review), 2026.

J. M. H. Pinheiro, G. N. Herrera, A. D. Santos, L. B. R. Fernandes, **R. V. Godoy**, E. A. B. Almeida, H. C. Onody, M. A. C. Vieira, A. M. Penteado-Dias, and M. Becker, "Automated Identification of Ichneumonoidea Wasps via YOLO-Based Deep Learning: Integrating HiresCam for Explainable AI", in *IEEE Access* (under review), 2026.

B. Guan, **R. V. Godoy**, and M. Liarokapis, "On the Impact of Different Light Wavelengths in Decoding Grip Forces for Lightmyography Controlled Prosthetic Hands", in *IEEE Access* (under review), 2026.

G. Tormena, D. Araújo, G. Carvalho, G. Centenaro, R. Pozzer, R. Kurosawa, D. Alves, F. Campos, P. Santos, P. Mota, **R. V. Godoy**, J. Pinheiro, and M. Becker, "Biological Sex Determination in Cadavers Using Deep Learning Algorithms from Computed Tomography Images of Pelvis and Skull", in *IEEE Access* (under review), 2026.

J. M. H. Pinheiro, S. V. B. de Oliveira, T. H. Segreto, P. A. R. Saraiva, E. F. de Souza, **R. V. Godoy**, L. A. Ambrosio, and M. Becker, "The Impact of Feature Scaling In Machine Learning: Effects on Regression and Classification Tasks", in *IEEE Access*, 2025.

B. Guan, **R. V. Godoy**, M. Shahmohammadi, A. Dwivedi, and M. Liarokapis, "Offline Versus Real-Time Grasp Prediction Employing a Wearable High-Density Lightmyography Armband: On the Control of Prosthetic Hands", in *IEEE Access*, 2025.

J. Buzzatto, H. Jiang, J. Liang, B. Busby, A. Lynch, **R. V. Godoy**, S. Matsunaga, R. Haraguchi, T. Mariyama, B. A. Macdonald, M. Liarokapis, "Multi-Layer, Sensorised Kirigami Grippers for Delicate yet Robust Robot Grasping and Single-Grasp Object Identification", in *IEEE Access*, 2024.

R. V. Godoy, B. Guan, F. Sanches, A. Dwivedi and M. Liarokapis, "Electromyography Based Gesture Decoding Employing Few-Shot Learning, Transfer Learning, and Training From Scratch", in *IEEE Access*, 2023.

M. Shahmohammadi, B. Guan, **R. V. Godoy**, A. Dwivedi, P. Nielsen, and M. Liarokapis, "On lightmyography based muscle-machine interfaces for the efficient decoding of human gestures and forces", in *Scientific Reports*, 2023.

R. V. Godoy et al., "Electromyography-Based, Robust Hand Motion Classification Employing Temporal Multi-Channel Vision Transformers", in *IEEE Robotics and Automation Letters (RA-L)*, 2022.

R. V. Godoy, A. Dwivedi and M. Liarokapis, "Electromyography Based Decoding of Dexterous, In-Hand Manipulation Motions With Temporal Multichannel Vision Transformers", in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2022.

R. V. Godoy et al., "On EMG Based Dexterous Robotic Telemanipulation: Assessing Machine Learning Techniques, Feature Extraction Methods, and Shared Control Schemes", in *IEEE Access*, 2022.

Conference Publications

F. Tommaselli, **R. V. Godoy**, T. Segreto, J Negri, and M. Becker, "Mind the Phase: Effective Rank and Representation Health in Legged Locomotion", in *Conference on Robot Learning (CoRL)*, 2026.

D. Almeida, J. Negri, G. Lazzarini, T. H. Segreto, R. Bezerra, **R. V. Godoy**, and M. Becker, "Language-Guided Grasping under Partial Observation for Mobile Manipulation in Field Inspection and Maintenance", in *IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, 2026.

L. K. de Oliveira, F. A. G. Tommaselli, J. A. Marsicano, M. S. Tayar, P. A. R. Saraiva, **R. V. Godoy**, M. Becker, "MIRA: A Modular Open-Source Micro-UAV for Indoor Research", in *IEEE Latin American Robotics Symposium (LARS)*, 2026.

S. M. C. Vanegas, D. K. P. Melo, M. A. Pastrana, **R. V. Godoy**, D. M. Muñoz, M. Becker, Input-Contract-Aware Conditioning for Reproducible Volumetric Mapping Across Heterogeneous Robotic Platforms, in *IEEE Latin American Robotics Symposium (LARS)*, 2026.

M. S. Taya, F. A. G. Tommaselli, G. Capezutto, P. A. R. Saraiva, P. H. V. de Freitas, L. Kido, G. Sonego, **R. V. Godoy**, and M. Becker, "VLN on the Fly: A Fully Onboard Vision-Language Navigation Stack for Aerial Robots", in *International Micro Air Vehicle conference and competition (IMAV)*, 2026.

G. Capezzuto, F. A. G. Tommaselli, M. P. Angarola, **R. V. Godoy**, and M. Becker, "Towards Capability-Aware Traversability Navigation for Unstructured Environments", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

R. R. Baptista, A. L. Salgado, **R. V. Godoy**, M. Becker, T. Boaventura, and G. J. G. Lahr, "Evaluating Zero-Shot and One-Shot Adaptation of Small Language Models in Leader-Follower Interaction", in *IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, 2026.

B. Guan, **R. V. Godoy**, A. Dwivedi, and M. Liarokapis, "On the Impact of Different Light Wavelengths in Decoding Human Intention in Lightmyography Controlled Prosthetic Hands", in 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2025.

B. Guan, M. Kobayashi, **R. V. Godoy**, M. Owen, and M. Liarokapis, "On Semi-Autonomous, Intuitive, Lightmyography Based Control of Humanlike Robotic and Prosthetic Hands Utilizing Video and IMU Data", in *IEEE International Conference on BioInformatics and BioEngineering (BIBE)*, 2025.

R. R. Baptista, N. R. Gerszberg, **R. V. Godoy**, and G. J. G. Lahr, "MIHRaGe: A Mixed-Reality Interface for Human-Robot Interaction via Gaze-Oriented Control", in *IEEE International Conference on Advanced Robotics (ICAR)*, 2025.

M. V. da Silva, M. H. Carvalho, J. Negri, T. H. Segreto, **R. V. Godoy**, and M. Becker, "A Vision-Based Shared-Control Teleoperation Scheme for Controlling the Robotic Arm of a Four-Legged Robot", **awarded the best paper** in the *IEEE Latin American Robotics Symposium (LARS)*, 2025.

M. S. Tayar, L. K. de Oliveira, J. Negri, T. H. Segreto, **R. V. Godoy**, and M. Becker, "Autonomous UAV Flight Navigation in Confined Spaces: A Reinforcement Learning Approach", in *IEEE Latin American Robotics Symposium (LARS)*, 2025.

P. Saraiva, E. Ferreira, J. Pinheiro, T. H. Segreto, **R. V. Godoy**, and M. Becker, "A Synthetic Dataset for Manometry Recognition in Robotic Applications", in *IEEE Latin American Robotics Symposium (LARS)*, 2025.

D. Almeida, G. Lazzarini, J. Negri, T. H. Segreto, **R. V. Godoy**, and M. Becker, "Optimizing Grasping in Legged Robots: A Deep Learning Approach to Loco-Manipulation", in *IEEE Latin American Robotics Symposium (LARS)*, 2025.

R. V. Godoy, B. Guan, A. Dwivedi, M. Owen, and M. Liarokapis, "A Video Dataset of Everyday Life Grasps for the Training of Shared Control Operation Models for Myoelectric Prosthetic Hands", in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024.

R. V. Godoy, B. Guan, A. Dwivedi, and M. Liarokapis, "An Affordances and Electromyography Based Telemanipulation Framework for Control of Robotic Arm-Hand Systems", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

R. V. Godoy, B. Guan, A. Dwivedi, M. Shahmohammadi, M. Owen, and M. Liarokapis, "Multi-Grasp Classification for the Control of Robot Hands Employing Transformers and Lightmyography Signals", in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2023.

B. Guan, **R. V. Godoy**, F. Sanches, A. Dwivedi, and M. Liarokapis, "On Semi-Autonomous Robotic Telemanipulation Employing Electromyography Based Motion Decoding and Potential Fields", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

M. Shahmohammadi, B. Guan, **R. V. Godoy**, and M. Liarokapis, "An Adaptive, Humanlike Prosthetic Hand Equipped with a Series Elastic Differential and a Lightmyography Based Control Interface", in *IEEE International Conference on Automation Science and Engineering (CASE)*, 2023.

B. Guan, **R. V. Godoy**, F. Sanches, A. Dwivedi, Y. Kwon, and M. Liarokapis, "Electromyography and Potential Fields Based Shared Control Framework for Robotic Telemanipulation", in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

N. Elangovan, **R. V. Godoy**, F. Sanches, K. Wang, T. White, P. Jarvis, and M. Liarokapis, "On Human Grasping and Manipulation in Kitchens: Automated Annotation, Insights, and Metrics for Effective Data Collection", in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

J. Liang, J. Buzzatto, B. Busby, **R. V. Godoy**, S. Matsunaga, R. Haraguchi, T. Mariyama, B. Macdonald, M. Liarokapis, "Employing Multi-Layer, Sensorised Kirigami Grippers for Single-Grasp Based Identification of Objects and Force Exertion Estimation", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

F. Sanches, G. Gao, N. Elangovan, **R. V. Godoy**, J. Chapman, K. Wang, P. Jarvis, M. Liarokapis, "Scalable, Intuitive Human to Robot Skill Transfer with Wearable Human Machine Interfaces: On Complex, Dexterous Tasks", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

R. V. Godoy et al., "Electromyography-Based, Robust Hand Motion Classification Employing Temporal Multi-Channel Vision Transformers", in *IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, 2022.

R. V. Godoy, A. Dwivedi, M. Shahmohammadi and M. Liarokapis, "Lightmyography Based Decoding of Human Intention Using Temporal Multi-Channel Transformers", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

N. Elangovan, C. Chang, **R. V. Godoy**, F. Sanches, K. Wang, P. Jarvis, and M. Liarokapis, "Comparing Human and Robot Performance in the Execution of Kitchen Tasks: Evaluating Grasping and Dexterous Manipulation Skills", in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2022.

R. V. Godoy et al., "Redundant Robot Kinematics Error Analysis for Neurosurgical Procedures", in *IEEE International Conference on Industry Applications (INDUSCON)*, 2021.

L. A. Marão, L. Casteluci, **R. V. Godoy**, H. Garcia, D. V. Magalhães and G. Caurin, "Deep Reinforcement Learning Control of an Autonomous Wheeled Robot in a Challenge Task: Combined Visual and Dynamics Sensoring", in *International Conference on Advanced Robotics (ICAR)*, 2019.

Preprint Publications

G. J. Lahr, **R. V. Godoy**, T. H. Segreto, J. O. Savazzi, A. Ajoudani, T. Boaventura, and G. A. Caurin, "Improving Failure Prediction in Aircraft Fastener Assembly Using Synthetic Data in Imbalanced Datasets", in *arXiv*, 2025.

R. V. Godoy et al., "EEG-Based Epileptic Seizure Prediction Using Temporal Multi-Channel Transformers", in *arXiv*, 2022.